Rajalakshmi Engineering College

Department of Computer Science and Design

GE23627 DESIGN THINKING AND INNOVATION

AY 2026-2027

S.C.O.P.E.S – Team Scope Sheet

Scoping of design challenge When you want to work on real project using Design Thinking, begin the process with a specific and purposeful situation or problem to solve called Design Challenge. The Design Challenge should be doable, clear, actionable and hopefully inspiring. Your problem statement should be an outcome and not a solution.

You can use SCOPES as a tool to brainstorm and frame your problem.

TEAM NAME	**MUSIQ**
TEAM MEMBERS	**MAHATHI B 241701029** **SNEHA S 241701055**
DOMAIN NAME	**Music Technology / MusicTech / Sound design**
DATE	**06/08/2026**
FACULTY INCHARGE	**Dr.P Revathy**

S – Situation / Problem • What is the real situation you want to address? • Where does this problem occur? • Describe without mentioning any solution.	Beginner and intermediate singers often struggle to practice singing effectively because they do not know their comfortable vocal range, suitable pitch, breathing capacity, or how to approach complex musical phrases such as sangathis. When practicing songs, they may sing in an unsuitable pitch, struggle with difficult sections, and lack clear feedback on what they need to improve. Existing karaoke platforms primarily focus on entertainment, while structured vocal-learning tools often require musical knowledge or professional guidance.
C – Constraints What are the CONSTRAINTS when trying to address the situation or problem?	• Users have different vocal ranges and abilities. • Beginners may not understand musical notes, pitch or sangathis. • Accurate pitch and vocal-range detection can be technically challenging. • Background music can affect voice analysis. • Advanced audio separation and AI features require significant processing. • Limited time, technical knowledge and resources for development.

O – Objectives & Outcomes What are the OBJECTIVES and intended OUTCOME of this design challenge?	**Objectives** • Understand the challenges faced by beginner singers. • Identify individual vocal range and capabilities. • Provide personalized singing practice and karaoke. • Simplify complex song sections for beginners. **Intended Outcomes** • Personalized exercises and song recommendations. • Easier understanding and practice of sangathis. • Improved pitch and breathing control. • Performance analysis and personalized improvement suggestions.
P – People Who are the PEOPLE that you are trying to help? Whose problem are you trying to address? Primary Stakeholders Secondary Stakeholders Decision Makers	**Primary Stakeholders** • Beginner singers • Amateur singers • Music students **Secondary Stakeholders** • Vocal teachers • Music academies • Karaoke enthusiasts **Decision Makers** • Music-learning platforms • Music academies • Educational institutions

E – Estimates What are the ESTIMATED resources required for this design challenge?	• Laptop / Microphone / Headphones • Figma (free) • Python & audio-processing libraries (free) • React / web development tools • AI / audio-processing APIs • GitHub (free) • Estimated Prototype Cost: **₹0–₹5,000**
S – Scope to Explore Is there ample SCOPE to explore multiple alternatives to address the problem?	• Personalized vocal-range detection • AI-generated alankars • Breathing exercises • Pitch & performance analysis • Sangathi / note visualization • Tempo slowing & looping • Pitch/key adjustment • Vocal & instrumental stem separation • Personalized song recommendations • AI-based vocal coaching
Frame the Design Challenge	How might we make singing practice and karaoke more personalized and accessible for beginners by adapting music, exercises and feedback to each user's vocal ability?

SIGNATURE